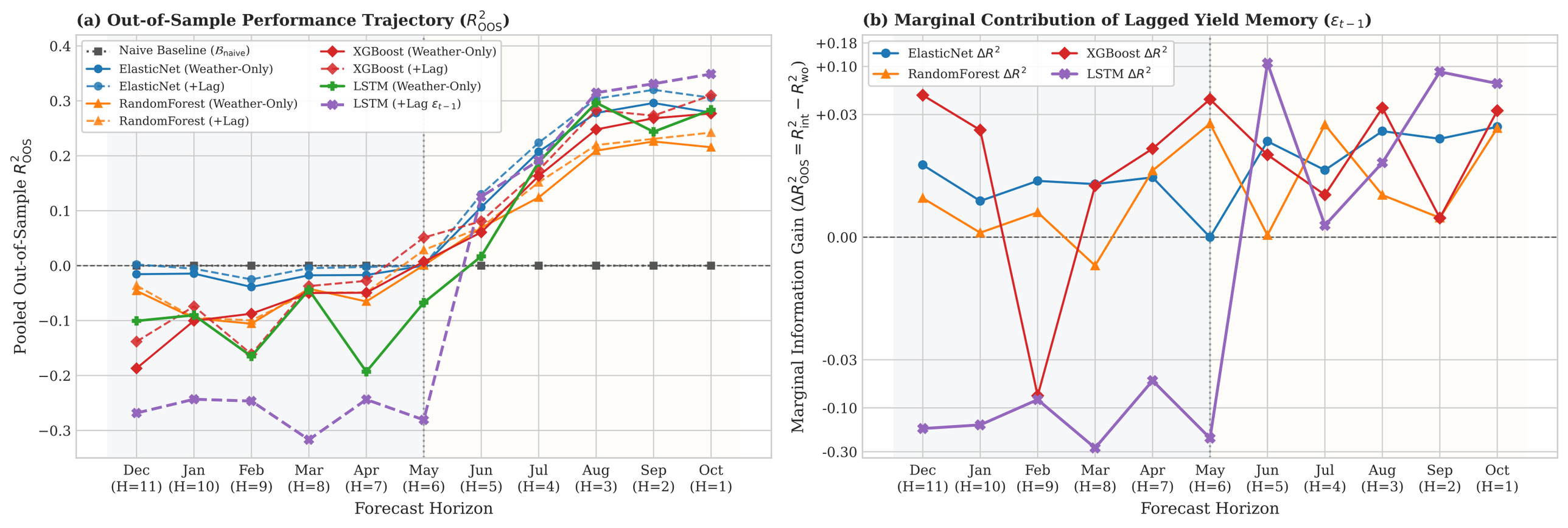
(a) Out-of-Sample Performance Trajectory (R^2_{OOS})
Naive Baseline ($\mathcal{B}_{naive}$)
ElasticNet (Weather-Only)
ElasticNet (+Lag)
RandomForest (Weather-Only)
RandomForest (+Lag)
XGBoost (Weather-Only)
XGBoost (+Lag)
LSTM (Weather-Only)
LSTM (+Lag ε_{t-1})
Pooled Out-of-Sample R^2_{OOS}
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Dec (H=11)
Jan (H=10)
Feb (H=9)
Mar (H=8)
Apr (H=7)
May (H=6)
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct (H=1)
Forecast Horizon
(b) Marginal Contribution of Lagged Yield Memory (ε_{t-1})
ElasticNet ΔR^2
RandomForest ΔR^2
XGBoost ΔR^2
LSTM ΔR^2
Marginal Information Gain ($\Delta R^2_{OOS} = R^2_{int} - R^2_{wo}$)
+0.18
+0.10
+0.03
0.00
-0.03
-0.10
-0.30
Dec (H=11)
Jan (H=10)
Feb (H=9)
Mar (H=8)
Apr (H=7)
May (H=6)
Jun (H=5)
Jul (H=4)
Aug (H=3)
Sep (H=2)
Oct (H=1)
Forecast Horizon